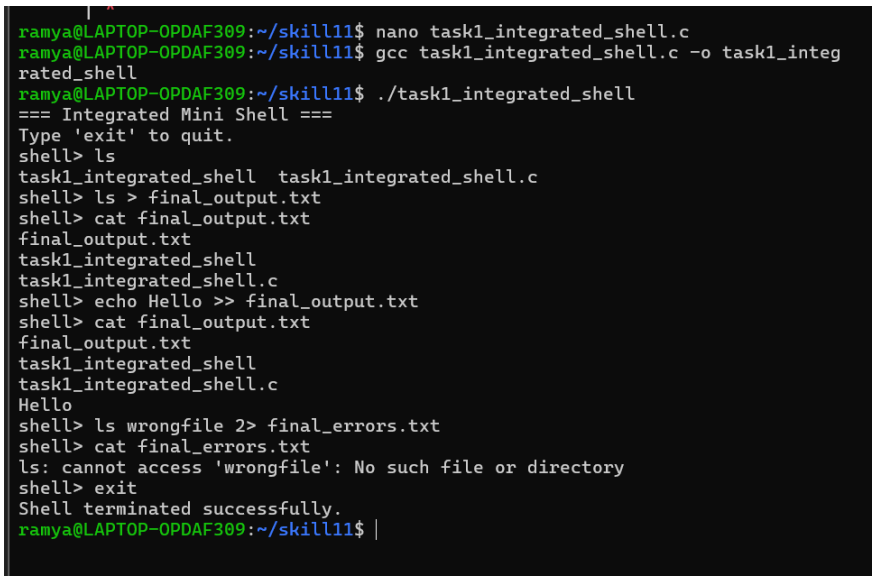
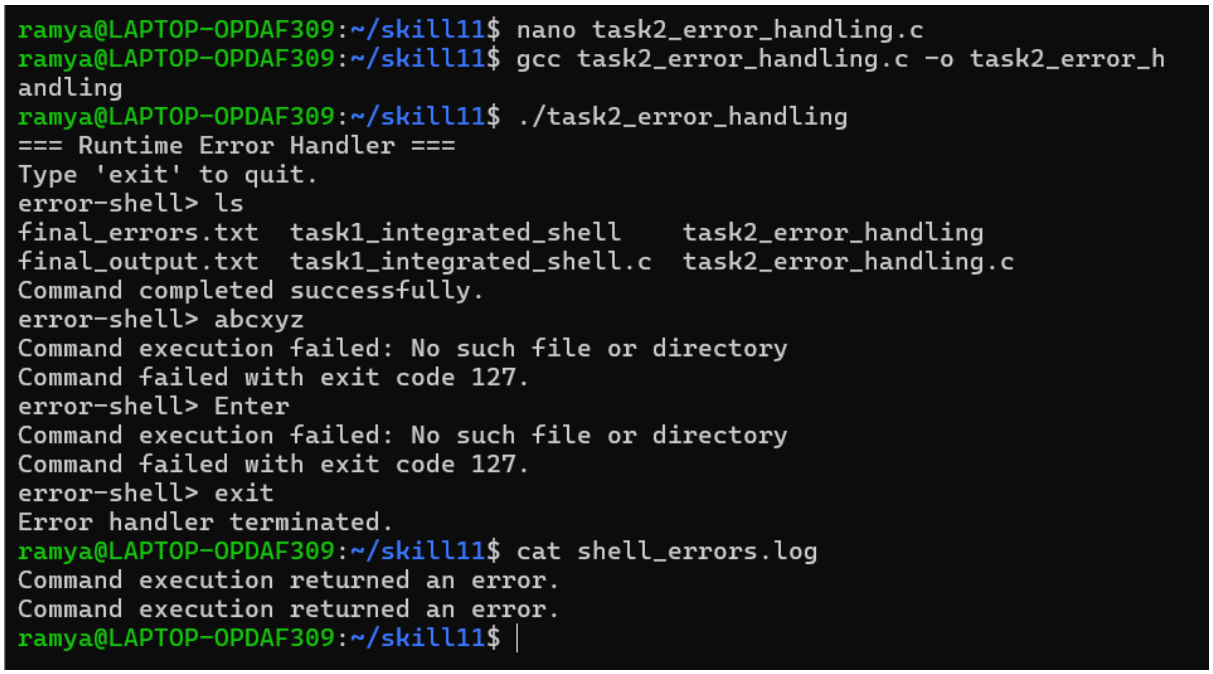
WEEK – 11

TASK 1 : To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reli ability, Refactor 4 Components



TASK 2 : TO Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios



TASK 3 : To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings

```
ramya@LAPTOP-OPDAF309:~/skill11$ nano task3_memory.c
ramya@LAPTOP-OPDAF309:~/skill11$ gcc -g task3_memory.c -o task3_memory
ramya@LAPTOP-OPDAF309:~/skill11$ ./task3_memory
Name: Linux Shell
Message: Memory management test completed.
Memory released successfully.
ramya@LAPTOP-OPDAF309:~/skill11$ valgrind --version
valgrind-3.26.0
ramya@LAPTOP-OPDAF309:~/skill11$ valgrind --leak-check=full --track-origins=
yes ./task3_memory
==3451== Memcheck, a memory error detector
==3451== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3451== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3451== Command: ./task3_memory
==3451==
Name: Linux Shell
Message: Memory management test completed.
Memory released successfully.
==3451==
==3451== HEAP SUMMARY:
==3451==     in use at exit: 0 bytes in 0 blocks
==3451==   total heap usage: 3 allocs, 3 frees, 1,174 bytes allocated
==3451==
==3451== All heap blocks were freed -- no leaks are possible
==3451==
==3451== For lists of detected and suppressed errors, rerun with: -s
==3451== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
ramya@LAPTOP-OPDAF309:~/skill11$
```

TASK 4 : To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate reports

```
ramya@LAPTOP-OPDAF309:~/skill11$ nano task4_test.sh
ramya@LAPTOP-OPDAF309:~/skill11$ chmod +x task4_test.sh
ramya@LAPTOP-OPDAF309:~/skill11$ ./task4_test.sh
==================================
     SHELL PROJECT TEST REPORT
==================================

Test 1: Compilation
PASS: Task 1 compilation
PASS: Task 2 compilation
PASS: Task 3 compilation

Test 2: Memory Program
Name: Linux Shell
Message: Memory management test completed.
Memory released successfully.
PASS: Memory cleanup test

Test 3: Output Redirection
PASS: Test input created

Test 4: Executable Verification
PASS: Integrated shell executable
PASS: Error handler executable
PASS: Memory test executable

==================================
Testing completed.
==================================
ramya@LAPTOP-OPDAF309:~/skill11$ cat test_report.txt
==================================
     SHELL PROJECT TEST REPORT
==================================

Test 1: Compilation
PASS: Task 1 compilation
PASS: Task 2 compilation
PASS: Task 3 compilation

Test 2: Memory Program
Name: Linux Shell
```

```
Test 3: Output Redirection
PASS: Test input created

Test 4: Executable Verification
PASS: Integrated shell executable
PASS: Error handler executable
PASS: Memory test executable

==================================
Testing completed.
==================================
ramya@LAPTOP-OPDAF309:~/skill11$ cat test_report.txt
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     SHELL PROJECT TEST REPORT
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Test 1: Compilation
PASS: Task 1 compilation
PASS: Task 2 compilation
PASS: Task 3 compilation

Test 2: Memory Program
Name: Linux Shell
Message: Memory management test completed.
Memory released successfully.
PASS: Memory cleanup test

Test 3: Output Redirection
PASS: Test input created

Test 4: Executable Verification
PASS: Integrated shell executable
PASS: Error handler executable
PASS: Memory test executable

==================================
Testing completed.
==================================
ramya@LAPTOP-OPDAF309:~/skill11$
```